

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: data = pd.read_csv("C:/Users/chait/Downloads/googleplaystore_user_reviews.csv")
```

```
In [3]: data.head()
```

```
Out[3]:
```

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
0	10 Best Foods for You	I like eat delicious food. That's I'm cooking ...	Positive	1.00	0.533333
1	10 Best Foods for You	This help eating healthy exercise regular basis	Positive	0.25	0.288462
2	10 Best Foods for You	NaN	NaN	NaN	NaN
3	10 Best Foods for You	Works great especially going grocery store	Positive	0.40	0.875000
4	10 Best Foods for You	Best idea us	Positive	1.00	0.300000

```
In [4]: data.shape
```

```
Out[4]: (64295, 5)
```

```
In [5]: data.sample(4)
```

Out[5]:

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivi
19421	CBS - Full Episodes & Live TV	NaN	NaN	NaN	NaN
52409	Garden Coloring Book	It's good	Positive	0.7000	0.6000
10362	BBVA Spain	NaN	NaN	NaN	NaN
46461	Fishdom	Cute game. The issue I fact money grabbing gam...	Negative	-0.2375	0.5933

In [6]: `data.columns`Out[6]: `Index(['App', 'Translated_Review', 'Sentiment', 'Sentiment_Polarity', 'Sentiment_Subjectivity'], dtype='object')`In [7]: `data.duplicated().sum()`Out[7]: `np.int64(33616)`In [9]: `data.drop_duplicates(inplace = True)`In [21]: `data.head(50)`

Out[21]:

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
0	10 Best Foods for You	I like eat delicious food. That's I'm cooking ...	Positive	1.000000	0.533333
1	10 Best Foods for You	This help eating healthy exercise regular basis	Positive	0.250000	0.288462
2	10 Best Foods for You	NaN	NaN	NaN	NaN
3	10 Best Foods for You	Works great especially going grocery store	Positive	0.400000	0.875000
4	10 Best Foods for You	Best idea us	Positive	1.000000	0.300000
5	10 Best Foods for You	Best way	Positive	1.000000	0.300000
6	10 Best Foods for You	Amazing	Positive	0.600000	0.900000
8	10 Best Foods for You	Looking forward app,	Neutral	0.000000	0.000000
9	10 Best Foods for You	It helpful site ! It help foods get !	Neutral	0.000000	0.000000
10	10 Best Foods for You	good you.	Positive	0.700000	0.600000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
10	Best	Useful information			
11	Foods for You	The amount spelling errors ...	Positive	0.200000	0.100000
10	Best	Thank you! Great app!! Add arthritis, eyes, im...	Positive	0.750000	0.875000
12	Foods for You				
10	Best	Greatest ever Completely awesome maintain heal...	Positive	0.992188	0.866667
13	Foods for You				
10	Best	Good health..... Good health first priority.....	Positive	0.550000	0.511111
14	Foods for You				
10	Best	Health It's important world either life . thin...	Positive	0.450000	1.000000
16	Foods for You				
10	Best	Mrs sunita bhati I thankful developers,to make...	Positive	0.600000	0.666667
17	Foods for You				
10	Best	Very Useful in diabetes age 30. I need control...	Positive	0.295000	0.100000
18	Foods for You				
10	Best	One greatest apps.	Positive	1.000000	1.000000
19	Foods for You				
10	Best	good nice	Positive	0.650000	0.800000
20	Foods for You				
10	Best	Healthy Really helped	Positive	0.350000	0.350000
21	Foods for You				

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
22	10 Best Foods for You	God health	Neutral	0.000000	0.000000
23	10 Best Foods for You	HEALTH SHOULD ALWAYS BE TOP PRIORITY. !!. ON M...	Positive	0.781250	0.500000
24	10 Best Foods for You	An excellent A useful	Positive	0.650000	0.500000
25	10 Best Foods for You	I found lot wealth form health...	Neutral	0.000000	0.000000
26	10 Best Foods for You	Because I found important.	Positive	0.400000	1.000000
27	10 Best Foods for You	Healthy Eating	Positive	0.500000	0.500000
28	10 Best Foods for You	Very good Simply good	Positive	0.805000	0.690000
29	10 Best Foods for You	On test....	Neutral	0.000000	0.000000
30	10 Best Foods for You	Good.!!	Positive	1.000000	0.600000
31	10 Best Foods for You	Thanks advice. Downloaded Adobe reader still c...	Positive	0.200000	0.200000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
10	Best Foods for You	No recipe book Unable recipe book.	Negative	-0.500000	0.500000
32	Best Foods for You	Absolutely Fabulous Phenomenal	Positive	0.450000	0.750000
33	Best Foods for You	Great Love food	Positive	0.650000	0.675000
34	Best Foods for You	Excellent It really works	Positive	0.600000	0.600000
35	Best Foods for You	Best	Positive	1.000000	0.300000
36	Best Foods for You	WEALTH OF INFORMATION Very informative ... eas...	Positive	0.186616	0.356313
37	Best Foods for You	Not description good	Positive	0.700000	0.600000
38	Best Foods for You	Great app. Love	Positive	0.650000	0.675000
39	Best Foods for You	Good	Positive	0.700000	0.600000
40	Best Foods for You	Loved Very helpful	Positive	0.450000	0.550000
41	Best Foods for You				

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
42	10 Best Foods for You	Good healthy foods.	Positive	0.600000	0.550000
43	10 Best Foods for You	Waste time It needs internet time n ask calls ...	Negative	-0.200000	0.000000
44	10 Best Foods for You	Best thing I lost ten lbs. two months help tips	Positive	0.500000	0.150000
45	10 Best Foods for You	Ten best food promote good health Good health ...	Positive	0.443750	0.412500
46	10 Best Foods for You	Healthy Food Really good information eat healthy	Positive	0.566667	0.533333
47	10 Best Foods for You	My mom dad important thing This good share	Positive	0.550000	0.800000
48	10 Best Foods for You	A big thanks ds I got bst gd health	Positive	0.100000	0.150000
49	10 Best Foods for You	Lovely Best everyone needs	Positive	0.750000	0.525000
50	10 Best Foods for You	Love It I lost 6Pounds far 215 209	Positive	0.300000	0.800000
51	10 Best Foods for You	I like	Neutral	0.000000	0.000000

```
In [12]: data.shape
```

```
Out[12]: (30679, 5)
```

```
In [13]: data.isnull().sum()
```

```
Out[13]: App                                0  
Translated_Review                        987  
Sentiment                               982  
Sentiment_Polarity                       982  
Sentiment_Subjectivity                   982  
dtype: int64
```

```
In [18]: data['Sentiment'].value_counts()
```

```
Out[18]: Sentiment  
Positive    19015  
Negative     6321  
Neutral     4361  
Name: count, dtype: int64
```

```
In [19]: data['Sentiment'].isnull().sum()
```

```
Out[19]: np.int64(982)
```

```
In [25]: data.loc[360:410]
```


Out[25]:

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
360	1LINE – One Line with One Touch	gets 1* there's ad every single level restart,...	Negative	-0.157143	0.70476
361	1LINE – One Line with One Touch	The game interesting challenging. However, oft...	Positive	0.200000	0.63333
362	1LINE – One Line with One Touch	NaN	NaN	NaN	NaN
363	1LINE – One Line with One Touch	Wow ,i play 15 puzzle 1st round I remember puz...	Positive	0.133333	0.71666
364	1LINE – One Line with One Touch	Love game deleting I've waiting 3 months level...	Positive	0.059091	0.36363
365	1LINE – One Line with One Touch	Way many aggressive ads	Positive	0.500000	0.50000
366	1LINE – One Line with One Touch	amaze i'm addicted	Negative	-0.400000	0.60000
367	1LINE – One Line with One Touch	I like game	Negative	-0.400000	0.40000
369	1LINE – One Line with One Touch	I play game	Negative	-0.400000	0.40000
370	1LINE – One Line with One Touch	It's tooooo gud game	Negative	-0.400000	0.40000
371	1LINE – One Line with One Touch	Too many unskippable ads. Other than, great game.	Positive	0.193750	0.50625
372	1LINE – One Line with One Touch	Mind uses game	Negative	-0.400000	0.40000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
373	1LINE – One Line with One Touch	Too much ad	Positive	0.200000	0.200000
374	1LINE – One Line with One Touch	It great experience knowledge help game	Positive	0.200000	0.575000
375	1LINE – One Line with One Touch	Nice game	Positive	0.100000	0.700000
376	1LINE – One Line with One Touch	I really good game	Positive	0.150000	0.500000
377	1LINE – One Line with One Touch	I love game nice	Positive	0.233333	0.666666
378	1LINE – One Line with One Touch	It granite	Neutral	0.000000	0.000000
379	1LINE – One Line with One Touch	It confusing	Negative	-0.300000	0.400000
380	1LINE – One Line with One Touch	Its really fun	Positive	0.300000	0.200000
381	1LINE – One Line with One Touch	Is game beautiful	Positive	0.225000	0.700000
382	1LINE – One Line with One Touch	Guide	Neutral	0.000000	0.000000
383	1LINE – One Line with One Touch	SUPER	Positive	0.333333	0.666666
384	1LINE – One Line with One Touch	yoko	Neutral	0.000000	0.000000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
385	1LINE – One Line with One Touch	It's great	Positive	0.800000	0.750000
386	1LINE – One Line with One Touch	Very nice	Positive	0.780000	1.000000
387	1LINE – One Line with One Touch	Good	Positive	0.700000	0.600000
388	1LINE – One Line with One Touch	Its cool	Positive	0.350000	0.650000
389	1LINE – One Line with One Touch	Cool	Positive	0.350000	0.650000
390	1LINE – One Line with One Touch	Fantastic	Positive	0.400000	0.900000
391	1LINE – One Line with One Touch	Superb	Positive	1.000000	1.000000
392	1LINE – One Line with One Touch	Excellent	Positive	1.000000	1.000000
393	1LINE – One Line with One Touch	Nice	Positive	0.600000	1.000000
394	1LINE – One Line with One Touch	Love	Positive	0.500000	0.600000
395	1LINE – One Line with One Touch	Fun	Positive	0.300000	0.200000
396	1LINE – One Line with One Touch	Enjoy	Positive	0.400000	0.500000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectiviti
397	1LINE – One Line with One Touch	Love ❤️❤️❤️❤️❤️❤️❤️❤️	Positive	0.500000	0.600000
398	1LINE – One Line with One Touch	Good graphics ok ,WAY many ads ridiculous can'...	Positive	0.233333	0.520000
399	1LINE – One Line with One Touch	Terrible amount ads covering content making un...	Negative	-0.825000	0.975000
400	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	Like The Old Version Better Where Backspace Is...	Positive	0.300000	0.350000
401	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	I like far	Positive	0.100000	1.000000
402	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	It's really useful I really like	Positive	0.250000	0.100000
403	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	I can't send emojis without saying change inpu...	Neutral	0.000000	0.000000
404	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	BullSh*t.... When open apps, says sill never r...	Neutral	0.000000	0.366666
405	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	NaN	NaN	NaN	NaN
406	2018Emoji Keyboard 🤔	Good	Positive	0.700000	0.600000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
	Emoticons Lite - sticker&gif				
	2018Emoji Keyboard				
408	Emoticons Lite - sticker&gif	Easy use, best keyboard useful..	Positive	0.577778	0.37777
	2018Emoji Keyboard				
410	Emoticons Lite - sticker&gif	Fuzzy	Neutral	0.000000	0.00000

```
In [26]: data.dropna( subset = ['Translated_Review'] , inplace =True)
```

```
In [27]: data.isnull().sum()
```

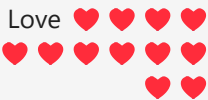
```
Out[27]: App                0
Translated_Review          0
Sentiment                  0
Sentiment_Polarity         0
Sentiment_Subjectivity     0
dtype: int64
```

```
In [28]: data.loc[360:410]
```

Out[28]:

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
360	1LINE – One Line with One Touch	gets 1* there's ad every single level restart,...	Negative	-0.157143	0.70476
361	1LINE – One Line with One Touch	The game interesting challenging. However, oft...	Positive	0.200000	0.63333
363	1LINE – One Line with One Touch	Wow ,i play 15 puzzle 1st round I remember puz...	Positive	0.133333	0.71666
364	1LINE – One Line with One Touch	Love game deleting I've waiting 3 months level...	Positive	0.059091	0.36363
365	1LINE – One Line with One Touch	Way many aggressive ads	Positive	0.500000	0.50000
366	1LINE – One Line with One Touch	amaze i'm addicted	Negative	-0.400000	0.60000
367	1LINE – One Line with One Touch	I like game	Negative	-0.400000	0.40000
369	1LINE – One Line with One Touch	I play game	Negative	-0.400000	0.40000
370	1LINE – One Line with One Touch	It's tooooo gud game	Negative	-0.400000	0.40000
371	1LINE – One Line with One Touch	Too many unskippable ads. Other than, great game.	Positive	0.193750	0.50625
372	1LINE – One Line with One Touch	Mind uses game	Negative	-0.400000	0.40000
373	1LINE – One Line with One Touch	Too much ad	Positive	0.200000	0.20000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
374	1LINE – One Line with One Touch	It great experience knowledge help game	Positive	0.200000	0.575000
375	1LINE – One Line with One Touch	Nice game	Positive	0.100000	0.700000
376	1LINE – One Line with One Touch	I really good game	Positive	0.150000	0.500000
377	1LINE – One Line with One Touch	I love game nice	Positive	0.233333	0.666666
378	1LINE – One Line with One Touch	It granite	Neutral	0.000000	0.000000
379	1LINE – One Line with One Touch	It confusing	Negative	-0.300000	0.400000
380	1LINE – One Line with One Touch	Its really fun	Positive	0.300000	0.200000
381	1LINE – One Line with One Touch	Is game beautiful	Positive	0.225000	0.700000
382	1LINE – One Line with One Touch	Guide	Neutral	0.000000	0.000000
383	1LINE – One Line with One Touch	SUPER	Positive	0.333333	0.666666
384	1LINE – One Line with One Touch	yoko	Neutral	0.000000	0.000000
385	1LINE – One Line with One Touch	It's great	Positive	0.800000	0.750000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
386	1LINE – One Line with One Touch	Very nice	Positive	0.780000	1.000000
387	1LINE – One Line with One Touch	Good	Positive	0.700000	0.600000
388	1LINE – One Line with One Touch	Its cool	Positive	0.350000	0.650000
389	1LINE – One Line with One Touch	Cool	Positive	0.350000	0.650000
390	1LINE – One Line with One Touch	Fantastic	Positive	0.400000	0.900000
391	1LINE – One Line with One Touch	Superb	Positive	1.000000	1.000000
392	1LINE – One Line with One Touch	Excellent	Positive	1.000000	1.000000
393	1LINE – One Line with One Touch	Nice	Positive	0.600000	1.000000
394	1LINE – One Line with One Touch	Love	Positive	0.500000	0.600000
395	1LINE – One Line with One Touch	Fun	Positive	0.300000	0.200000
396	1LINE – One Line with One Touch	Enjoy	Positive	0.400000	0.500000
397	1LINE – One Line with One Touch	Love 	Positive	0.500000	0.600000

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectiviti
398	1LINE – One Line with One Touch	Good graphics ok ,WAY many ads ridiculous can'...	Positive	0.233333	0.52000
399	1LINE – One Line with One Touch	Terrible amount ads covering content making un...	Negative	-0.825000	0.97500
400	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	Like The Old Version Better Where Backspace Is...	Positive	0.300000	0.35000
401	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	I like far	Positive	0.100000	1.00000
402	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	It's really useful I really like	Positive	0.250000	0.10000
403	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	I can't send emojis without saying change inpu...	Neutral	0.000000	0.00000
404	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	BullSh*t.... When open apps, says sill never r...	Neutral	0.000000	0.36666
406	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	Good	Positive	0.700000	0.60000
408	2018Emoji Keyboard 🤔 Emoticons Lite - sticker&gif	Easy use, best keyboard useful..	Positive	0.577778	0.37777

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectiviti
410	2018Emoji Keyboard Emoticons Lite - sticker&gif	Fuzzy	Neutral	0.000000	0.000000

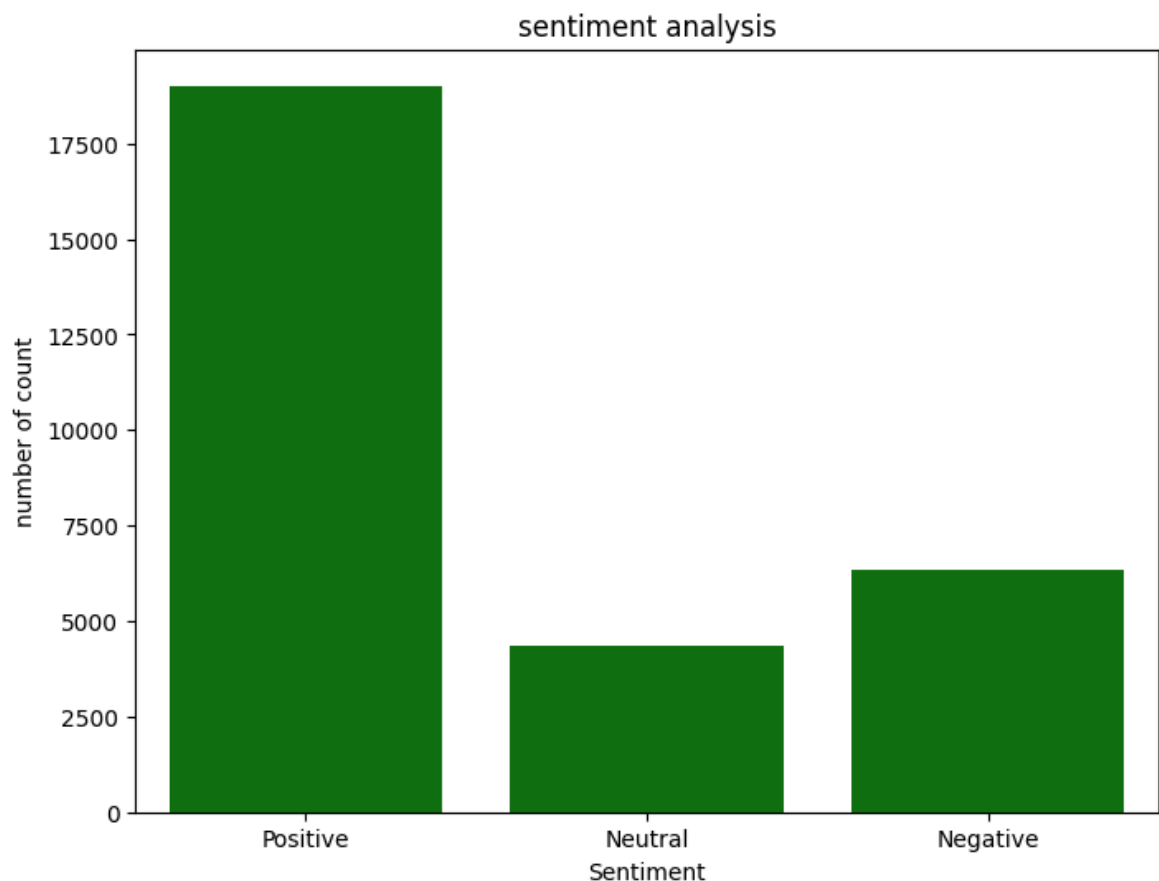
In [29]: `data.shape`

Out[29]: (29692, 5)

In [30]: `data.dtypes`

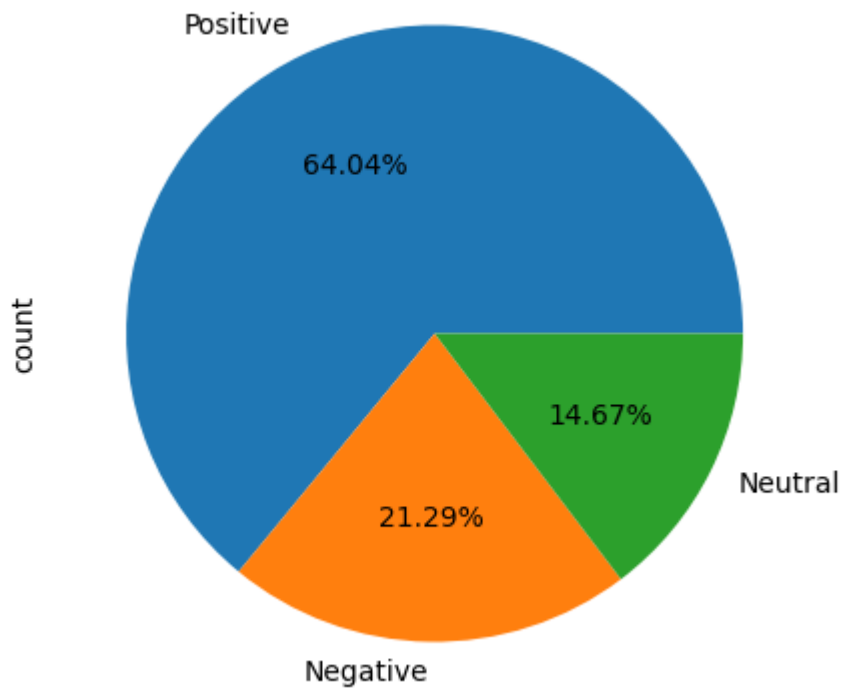
Out[30]: App object
Translated_Review object
Sentiment object
Sentiment_Polarity float64
Sentiment_Subjectivity float64
dtype: object

In [34]: `plt.figure(figsize = (8,6))
sns.countplot(data = data ,x="Sentiment" , color = "Green")
plt.title("sentiment analysis")
plt.ylabel('number of count')
plt.show()`

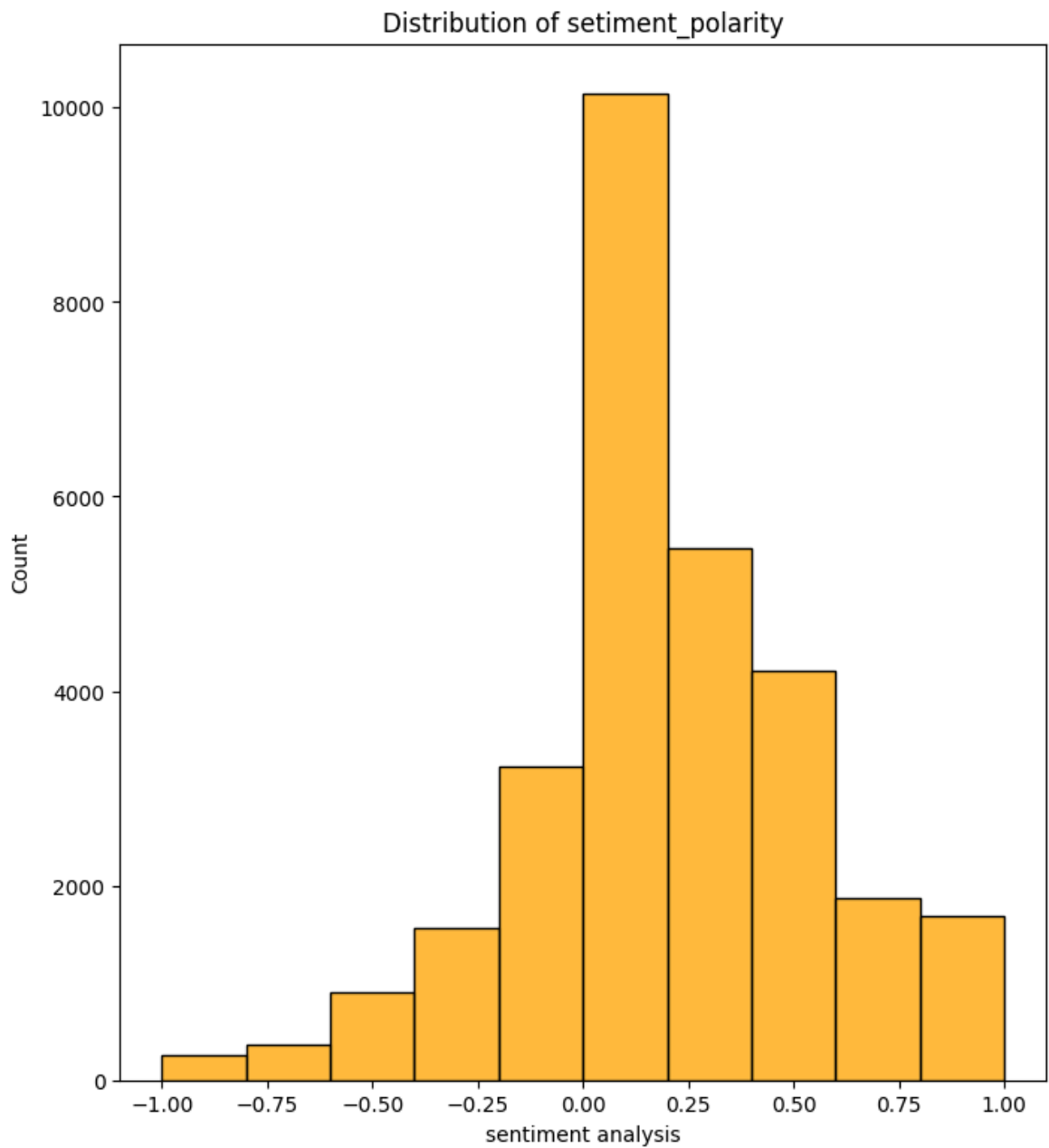


In [38]: `plt.figure(figsize = (7,5))
data['Sentiment'].value_counts().plot(kind = "pie", autopct = "%1.2f%%")`

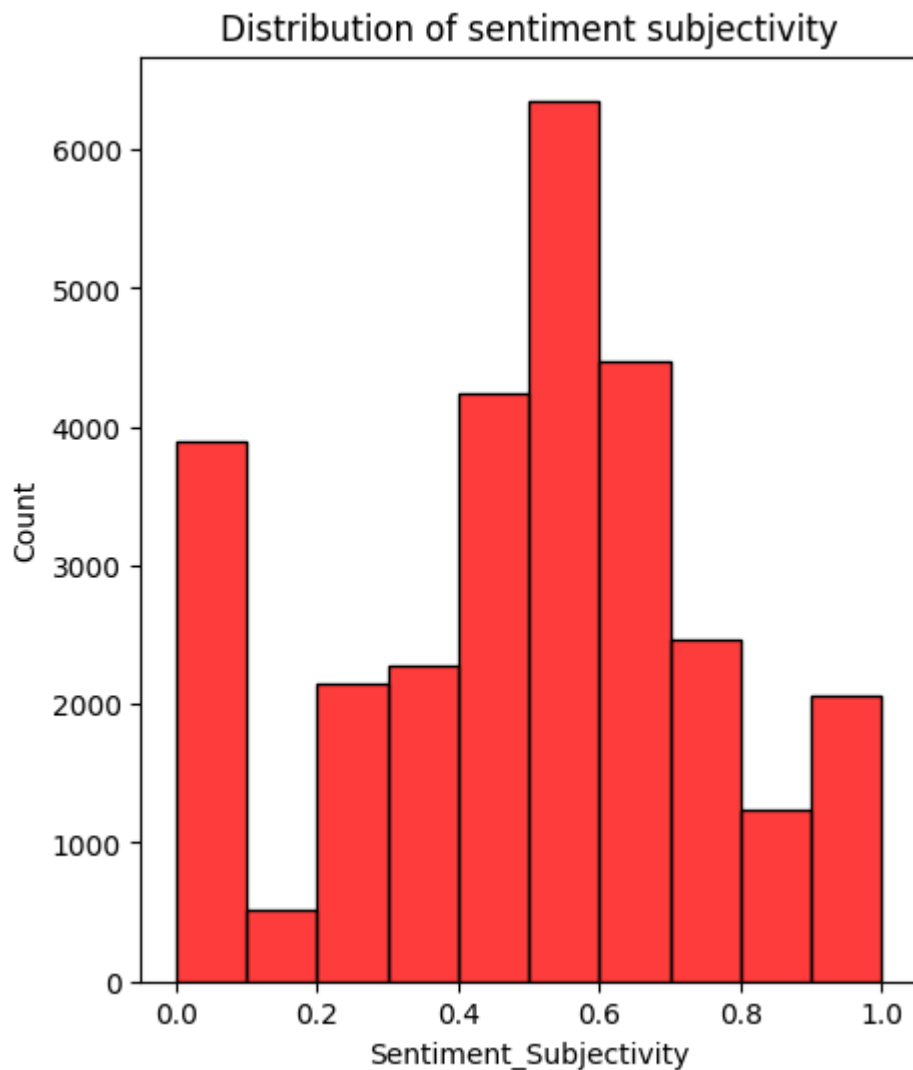
```
plt.show()
```



```
In [41]: plt.figure( figsize = (8,9))
sns.histplot( data['Sentiment_Polarity'] , color = 'orange' , bins = 10)
plt.title('Distribution of setiment_polarity')
plt.xlabel('sentiment analysis')
plt.show()
```



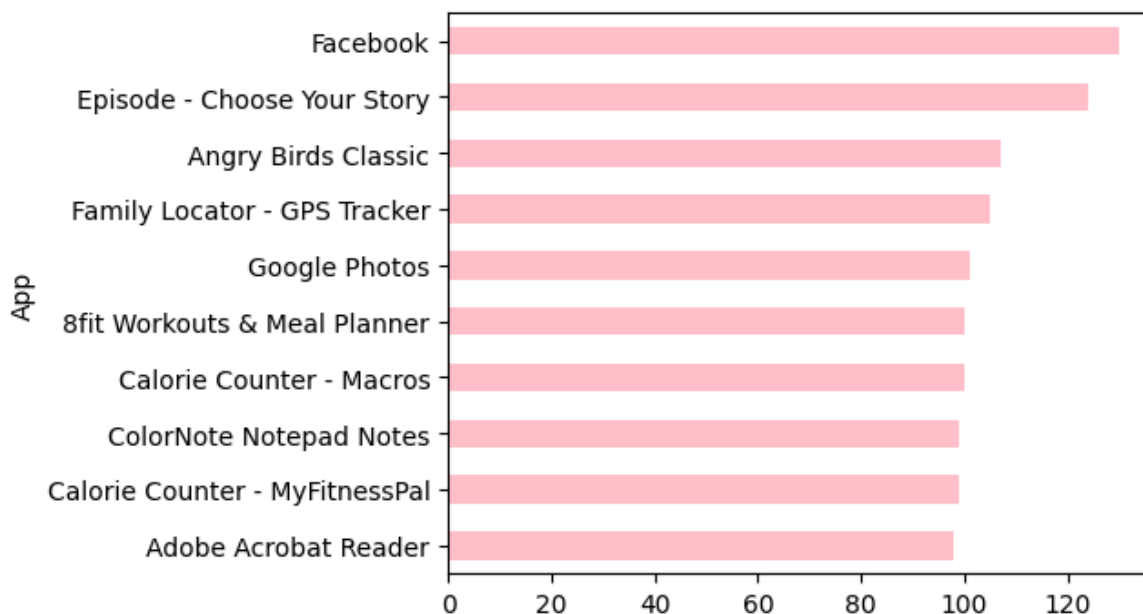
```
In [42]: plt.figure(figsize = (5,6))
sns.histplot(data['Sentiment_Subjectivity'] , bins = 10 , color = 'red')
plt.title('Distribution of sentiment subjectivity')
plt.show()
```



```
In [51]: top_app = data['App'].value_counts().head(10).sort_values()
         print(top_app)
```

```
App
Adobe Acrobat Reader           98
Calorie Counter - MyFitnessPal 99
ColorNote Notepad Notes        99
Calorie Counter - Macros       100
8fit Workouts & Meal Planner    100
Google Photos                  101
Family Locator - GPS Tracker    105
Angry Birds Classic            107
Episode - Choose Your Story     124
Facebook                       130
Name: count, dtype: int64
```

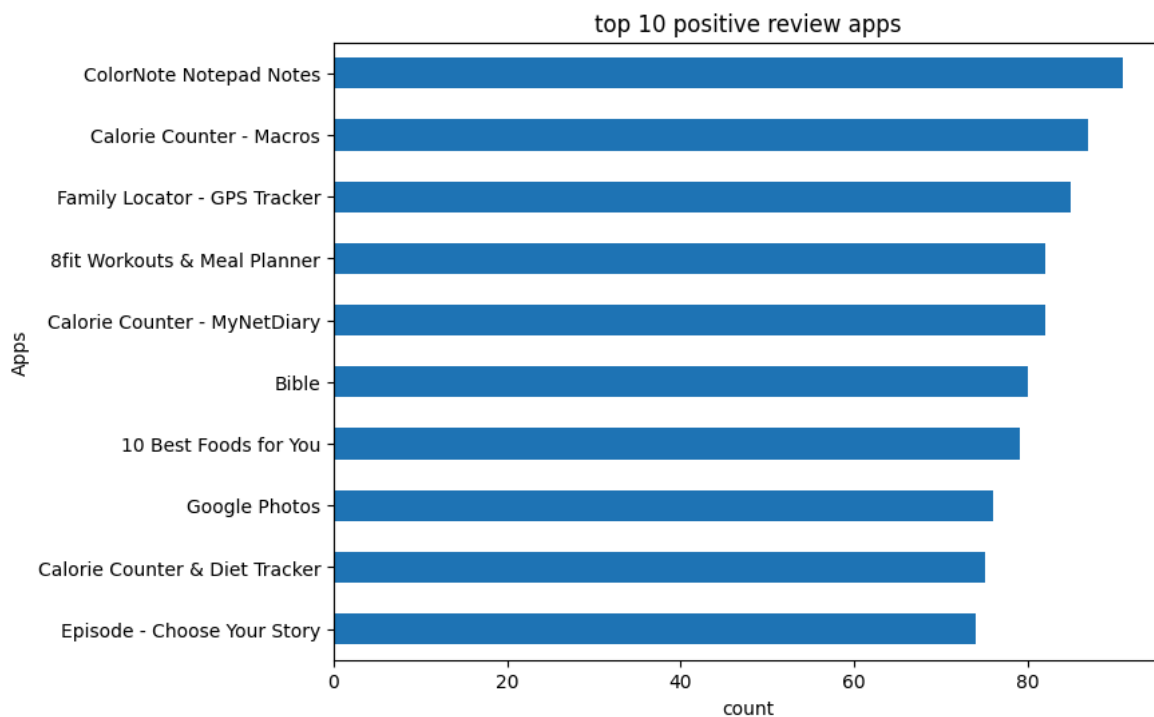
```
In [56]: plt.figure( figsize = (5,4))
         top_app.plot( kind = 'barh', color = 'Pink' )
         plt.show()
```



```
In [71]: top_positive_app = (data['App'][data['Sentiment'] == 'Positive'].value_counts().
print(top_positive_app)
```

```
App
Episode - Choose Your Story    74
Calorie Counter & Diet Tracker  75
Google Photos                 76
10 Best Foods for You         79
Bible                        80
Calorie Counter - MyNetDiary  82
8fit Workouts & Meal Planner  82
Family Locator - GPS Tracker  85
Calorie Counter - Macros     87
ColorNote Notepad Notes     91
Name: count, dtype: int64
```

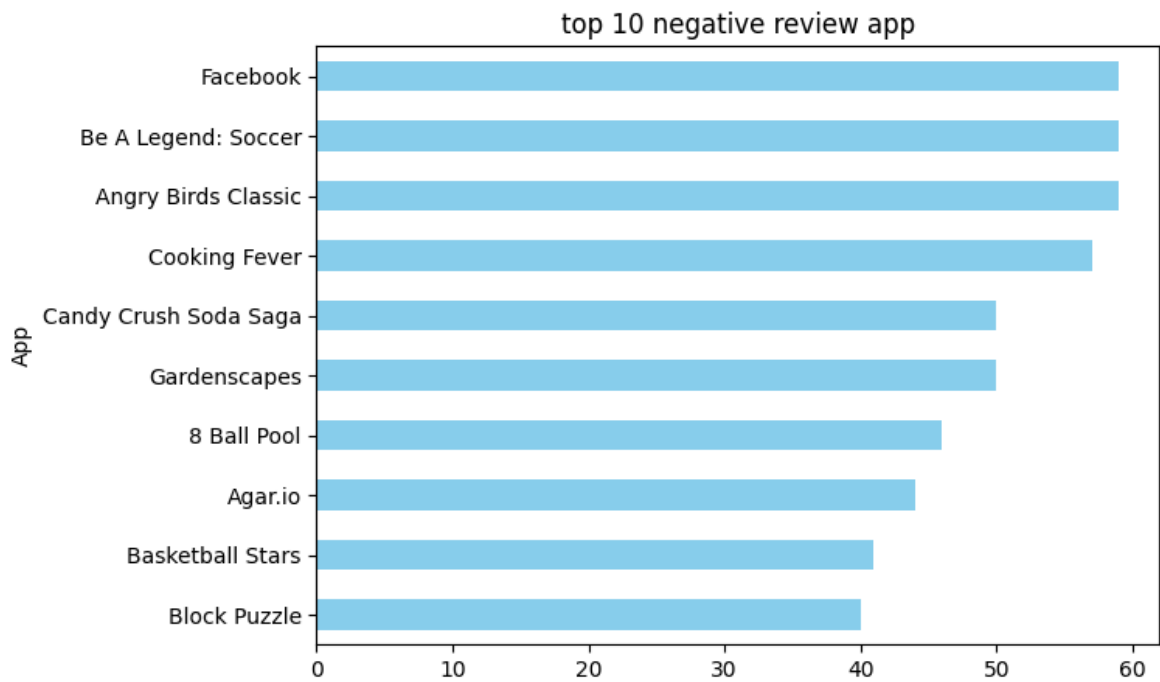
```
In [74]: plt.figure( figsize = (8,6))
top_positive_app.plot(kind = "barh")
plt.title('top 10 positive review apps')
plt.xlabel('count')
plt.ylabel('Apps')
plt.show()
```



```
In [81]: top_negative = data['App'][data['Sentiment'] == 'Negative'].value_counts().head(
print(top_negative)
```

```
App
Block Puzzle          40
Basketball Stars      41
Agar.io               44
8 Ball Pool           46
Gardenscapes          50
Candy Crush Soda Saga 50
Cooking Fever         57
Angry Birds Classic   59
Be A Legend: Soccer   59
Facebook              59
Name: count, dtype: int64
```

```
In [82]: plt.figure(figsize = (7,5))
top_negative.plot( kind = "barh" , color = "Skyblue")
plt.title("top 10 negative review app")
plt.show()
```



In [84]: `data.describe()`

Out[84]:

	Sentiment_Polarity	Sentiment_Subjectivity
--	--------------------	------------------------

count	29692.000000	29692.000000
mean	0.188868	0.490930
std	0.355694	0.265976
min	-1.000000	0.000000
25%	0.000000	0.350000
50%	0.157143	0.514286
75%	0.422917	0.652703
max	1.000000	1.000000

In [103...

```
avgsa = data.groupby('App')['Sentiment_Polarity'].mean().head(20).sort_values(ascending=True)
print(avgsa)
```



```

App
3D Live Neon Weed Launcher                                0.568182
10 Best Foods for You                                    0.465906
365Scores - Live Scores                                  0.438312
2018Emoji Keyboard 🤪 Emoticons Lite -sticker&gif      0.423659
2RedBeans                                                0.385160
104 找工作 - 找工作 找打工 找兼職 履歷健檢 履歷診療室    0.358227
30 Day Fitness Challenge - Workout at Home              0.333537
7 Cups: Anxiety & Stress Chat                            0.324670
1800 Contacts - Lens Store                              0.318145
4 in a Row                                                0.281144
2Date Dating App, Love and matching                     0.280267
21-Day Meditation Experience                             0.255183
2GIS: directory & navigator                              0.223129
591房屋交易-香港                                       0.222917
1LINE - One Line with One Touch                         0.196290
7 Day Food Journal Challenge                             0.189411
11st                                                      0.185943
4K Wallpapers and Ultra HD Backgrounds                  0.179328
2ndLine - Second Phone Number                           0.168968
591房屋交易-租屋、中古屋、新建案、實價登錄、別墅透天、公寓套房、捷運、買房賣房行情、房價
房貸查詢      0.080792
Name: Sentiment_Polarity, dtype: float64

```

```

In [108... avgss = data.groupby('App')['Sentiment_Subjectivity'].mean().sort_values(ascendi
print(avgss)

```

```

App
Google Slides                                             0.916667
Daily Workouts - Exercise Fitness Routine Trainer       0.750000
Choice Hotels                                             0.719206
HTC Calendar                                              0.694949
Hitwe - meet people and chat                             0.694105
850 Sports News Digest                                   0.692420
Google Street View                                       0.691667
CM Browser - Ad Blocker , Fast Download , Privacy       0.687500
Draw Your Game                                           0.683333
FlipaClip - Cartoon animation                           0.679226
Google Primer                                             0.675000
Aviary Stickers: Free Pack                               0.672514
A Word A Day                                              0.669034
Galactic Core Free Wallpaper                             0.657814
A Call From Santa Claus!                                0.657741
Call Blocker                                              0.655431
HelloTalk - Chat, Speak & Learn Foreign Languages       0.653056
Coloring book moana                                     0.645694
Cookbook Recipes                                         0.642612
365Scores - Live Scores                                  0.639610
Name: Sentiment_Subjectivity, dtype: float64

```

```

In [109... avgss = data.groupby('App')['Sentiment_Subjectivity'].mean().sort_values().head(
print(avgss)

```

App	
HD Camera	0.000000
CBS News	0.000000
Google Trips - Travel Planner	0.000000
Free Live Talk-Video Call	0.100000
Calculator - free calculator, multi calculator app	0.154167
Google Voice	0.196222
Apartment Decorating Ideas	0.200000
Acorns - Invest Spare Change	0.203889
Allegiant	0.233333
Calendar Widget Month + Agenda	0.250000
Flashlight & LED Torch	0.258178
HTC File Manager	0.261051
Free Panda Radio Music	0.266667
El tiempo de AEMET	0.275437
Call Control - Call Blocker	0.291667
HomeWork	0.300000
BeautyPlus - Easy Photo Editor & Selfie Camera	0.300000
ANA	0.300527
AdWords Express	0.301638
Caf - Mon Compte	0.303846
Name: Sentiment_Subjectivity, dtype: float64	

In []: